Final Report: Expected goals (xG)

Andy Song, Jake Summaria, Samy Bouzamondo, Yifeng Luo
Faculty Mentor: Kiran Luecke (Mathematics Department, kiranl@illinois.edu)
Graduate Mentor: Yigal Kamel

December 23, 2024

Abstract

We study some aspects of the expected Goals (xG) metric in soccer statistics. More specifically, we investigate xG trends in outlier goals (early goals, late goals, and game-winning goals), and investigate the dependence of xG on more detailed data than just field position (e.g. perceived goal area, number of defenders in line-of-sight, and shot-type).

1 Introduction

Compared to other popular sports such as basketball, American football, and baseball, the use of statistical methods in soccer is relatively new. The current state-of-the-art is a metric called "expected goals" (xG). In its most basic form, xG is a real-valued function of two variables, $F(x, y)$, which encodes the probability of scoring a goal when taking a shot from position (x, y) on the soccer field.

Our project has two parts: firstly, we use an existing xG function to study trends *within outlier goals*. Secondly we investigate whether the xG function can be refined to a better predictor of goal-scoring trends if one includes more detailed data than just the field position of the shot.

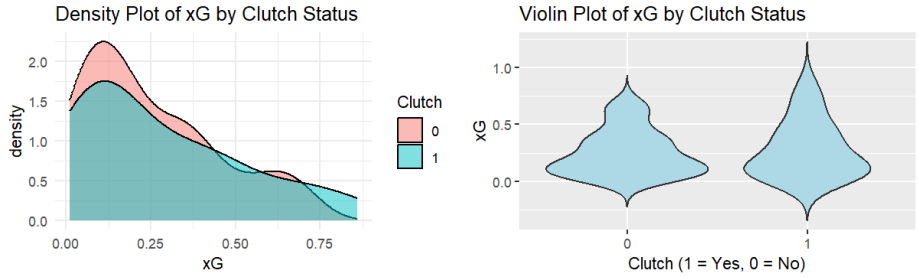
2 Results

2.1 xG trends within outliers

The first type of outlier goal we study are *clutch goals*, which we define to be those goals which are either scored after the 80th minute of the game or move the scoring team out of a losing score.

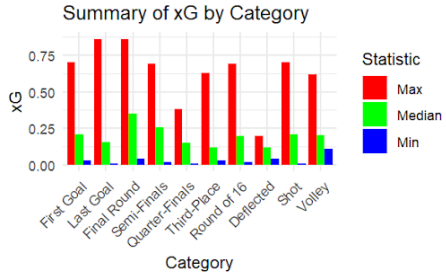
A *Chi-squared Test* was performed to assess whether the distribution of clutch goals across the game was significantly different from what would be expected by chance. The results ($X^2 = 49.838$, $p < 0.001$) revealed a strong association, indicating that clutch goals disproportionately occur in the final 10 minutes of gameplay.

Visualization of results:



To confirm these observations, a *Welch two-sample t-test* was performed, which returned a *t-statistic* of -0.49495 and a *p-value* of 0.6244. These results indicate that the difference in xG between clutch and non-clutch goals is not statistically significant.

The second analysis we performed was an investigation into the trends in xG in a tournament.

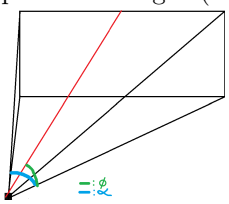


2.2 Refinements of xG

First we investigated the dependence of xG on the *perceived goal length*.

Definition 2.2.1. The perceived length is the visual angle of the goal as seen from the shooters perspective. We can extend this logic to 3D by including the height of the posts.

Below is a drawing of how the perceived Length 3d is computed. It is defined as the product of the angles α and ϕ which are determined by using trigonometric identities. It turned out that the perceived area (PA), as compared to the perceived length (PL), was not beneficial to the model.



Here is the formula for the percieved lengt (PL) and perceived area (PA):

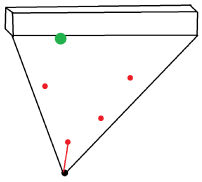
$$\text{PL} = \left(\frac{180}{\pi} \cdot \arccos \left(\frac{d^2_{\text{shot-to-left-post}} + d^2_{\text{shot-to-right-post}} - d^2_{\text{post-to-post}}}{2 \cdot d_{\text{shot-to-left-post}} \cdot d_{\text{shot-to-right-post}}} \right) \right)$$

$$\text{PA} = \text{PL} \cdot \left(\frac{180}{\pi} \cdot \arccos \left(\frac{d^2_{\text{top-to-shot}} + d^2_{\text{ground-mid-to-shot}} - d^2_{\text{top-to-mid}}}{2 \cdot d_{\text{top-to-shot}} \cdot d_{\text{ground-mid-to-shot}}} \right) \right)$$

Next we investigated the effect of the presence defenders between the shot taker and the goal, as well as the type of shot (header vs standard shot), and we used neural networks to tweak the coefficients in the resulting model.

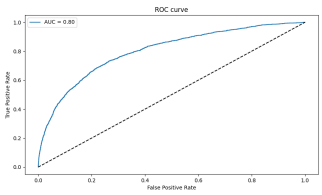
Definition 2.2.2. The *freeze frame* of a shot is the data of the coordinates of every player on the field at the time the shot was taken. The closest player in the triangle as in the definition of PL (formed by the shot location and the goal posts) had the greatest chance of affecting the outcome of a shot and was considered in the model.

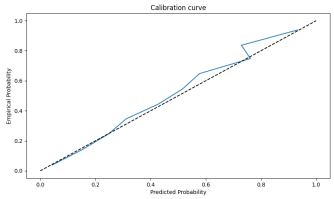
Here is a visual representation of a freeze frame: The distance between closest defender (points in red) and shot location is considered to have greatest impact on goal outcome. The goalkeeper position (point in green) is also considered in our model.



Our intuition was that the closest player in the triangle as defined in the 2d Perceived Length had the greatest chance of affecting the outcome of a shot. We also considered whether the shot was taken as a header, as this represents a fundamentally different type of attempt compared to shots taken with the foot.

Below are some graphics that summarize the neural network training.





2.3 Shot Location Analysis and xG Model

We analyzed data from StatsBomb's open format, comprising 3,433 matches with 87,111 total shots, of which 9,667 resulted in goals (11.1% conversion rate).

The goal was to develop a model analyzing shot location effectiveness in football, focusing on the relationship between shooting position, angle, and scoring probability.

First, we standardized the spatial variables used in our analysis.

Definition 2.3.1. The standardized coordinates of a shot are derived from the raw (x, y) coordinates, distance d, and angle θ through a normalization process based on the statistical distribution of these variables in our dataset.

The standardization formulas are:

$$x_{std} = \frac{x - 103.571}{8.802}, \quad y_{std} = \frac{y - 39.761}{9.895}$$

$$d_{std} = \frac{d - 19.203}{8.753}, \quad \theta_{std} = \frac{\theta - 0.443}{0.276}$$

These standardized variables are then used to compute the probability of scoring through our xG model:

$$\text{log odds} = -2.422 - 0.381x_{std} + 0.007y_{std} - 1.125d_{std} + 0.234\theta_{std}$$

$$\text{xG} = \frac{1}{1 + e^{-\text{log odds}}}$$

The model reveals several key relationships between spatial factors and scoring probability:

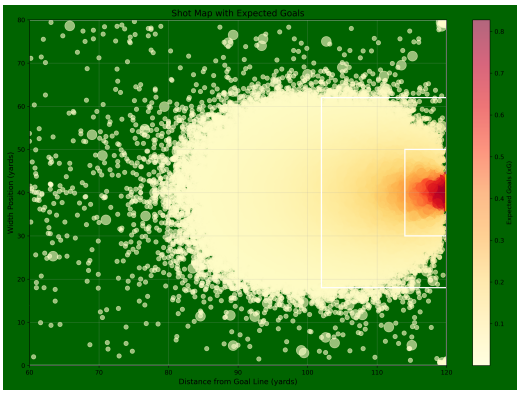
- Shot distance emerges as the strongest negative predictor (-1.125)
- Field position shows moderate negative influence (-0.381)
- Shot angle demonstrates positive impact (0.234)
- Width position exhibits minimal effect (0.007)

Definition 2.3.2. A high-probability scoring zone is defined as an area where the expected goals value exceeds 0.4, typically found within the 6-yard box and central penalty area.

Our analysis identified three distinct zones of heightened scoring probability:

- Shots within the 6-yard box show highest xG (0.4-0.8)
- Central penalty area shots average 0.2-0.4 xG
- Penalty spot location records peak xG values of $\geqslant 0.7$

Below is a visualization of the shot location heat map showing the xG distribution:



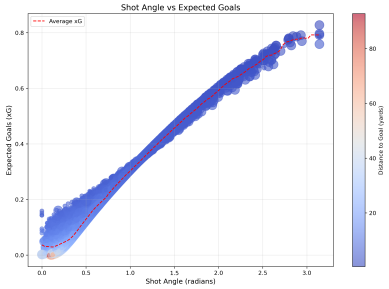
The color gradient represents scoring probability, with:

- Red regions indicating high xG ($\geqslant 0.4$)
- Green regions indicating low xG ($\leqslant 0.1$)

The white lines indicate the boundaries of the penalty area and 6-yard box.

Definition 2.3.3. The optimal shooting position is defined as the location that maximizes xG, typically found in central areas less than 12 yards from goal where both distance and angle are optimized.

The model was validated using an 80-20 train-test split (training: n=69,689, validation: n=17,422), achieving 76% prediction accuracy. Below are the model performance metrics:



The analysis supports several key tactical implications:

Definition 2.3.4. The tactical value of a shooting position is determined by its expected goals value, with positions being classified as high-value (xG $\geqslant$ $0.4), moderate - value(0.2 \leqslant xG \leqslant 0.4), orlow - value(xG \leqslant 0.2)$.

The data strongly supports tactical approaches prioritizing central penetration in attacking phases:

- High-value shooting positions strongly correlate with central penalty area
- Wide positions beyond penalty area show consistently low xG ($\leqslant 0.1)Scoringprobabilitydecreasesexponer$
- Wide angles significantly reduce xG regardless of distance
- Optimal shooting positions cluster in central areas $\leqslant 12yardsfromgoal$

3 Conclusion and future work

Our conclusions are as follows: 1) There were no significant trends in the clutch goals, despite a small increase in the average xG. 2) Considering known xG functions in terms of perceived goal length can significantly improve the metrics predictive power. Considering it as a function of perceived goal area does not increase its predictive power. Taking into account the freeze frame (in particular the distance of the nearest defender between the shot taker and the goal mouth) increases the predictive power. The use of neural network training to adjust parameters in our model was significantly beneficial.

Possible future work includes 1) a larger study of outliers, including tournament winning goals from a larger class of tournaments, 2) a study of the xG of own goals, 3) the addition of even more detailed data, for example the inclusion of the entire freeze frame, as opposed to just the nearest defender, 4) the dependence of player specific metrics, such as blood oxygen, player height, and shoe size.

4 References

1. https://t.co/qUxJ6GHHWE
2. https://www.americansocceranalysis.com/explanation
3. https://www.twenty3.sport/xg-what-is-it-explained/
4. https://tonyelhabr.rbind.io/posts/xg-predictor-future-results/
5. https://statsbomb.com/what-we-do/hub/free-data/
6. https://github.com/statsbomb/open-data
7. https://www.americansocceranalysis.com/home/2017/3/6/validating-the-asa-xgoals-model
8. https://fbref.com/en/